# DRILL

A study machine with no material in it

**Build 25** — sections, recall practice and a graded test that opens the next section when you pass it. It ships empty, and what you load into it never leaves your device.

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## 1. What this is, and why it is empty

DRILL is the machine, not the material. It holds no questions, no reference text and nothing of anybody’s. You load a **content pack** from your own device; it is read here in the browser, and it stays on the device it was loaded on.

That is not a limitation, it is the design:

1. **Material somebody else owns can be studied without ever being published.** The machine is published; the material is yours and stays yours.
2. **It makes the permission conversation a short one.** You can show somebody the machine running, empty, and say “your material drops in as a pack — here is what it looks like loaded”. If they say yes, publish the pack or hand it to them to maintain. If they say no, nothing needs undoing.
3. **Nothing is uploaded, ever.** The file is read by the page. There is no server to send it to.

This tool ships empty. It holds no questions and no material of its own — you load a content pack from your own device, it is read here in the browser, and it never leaves the device it is on.

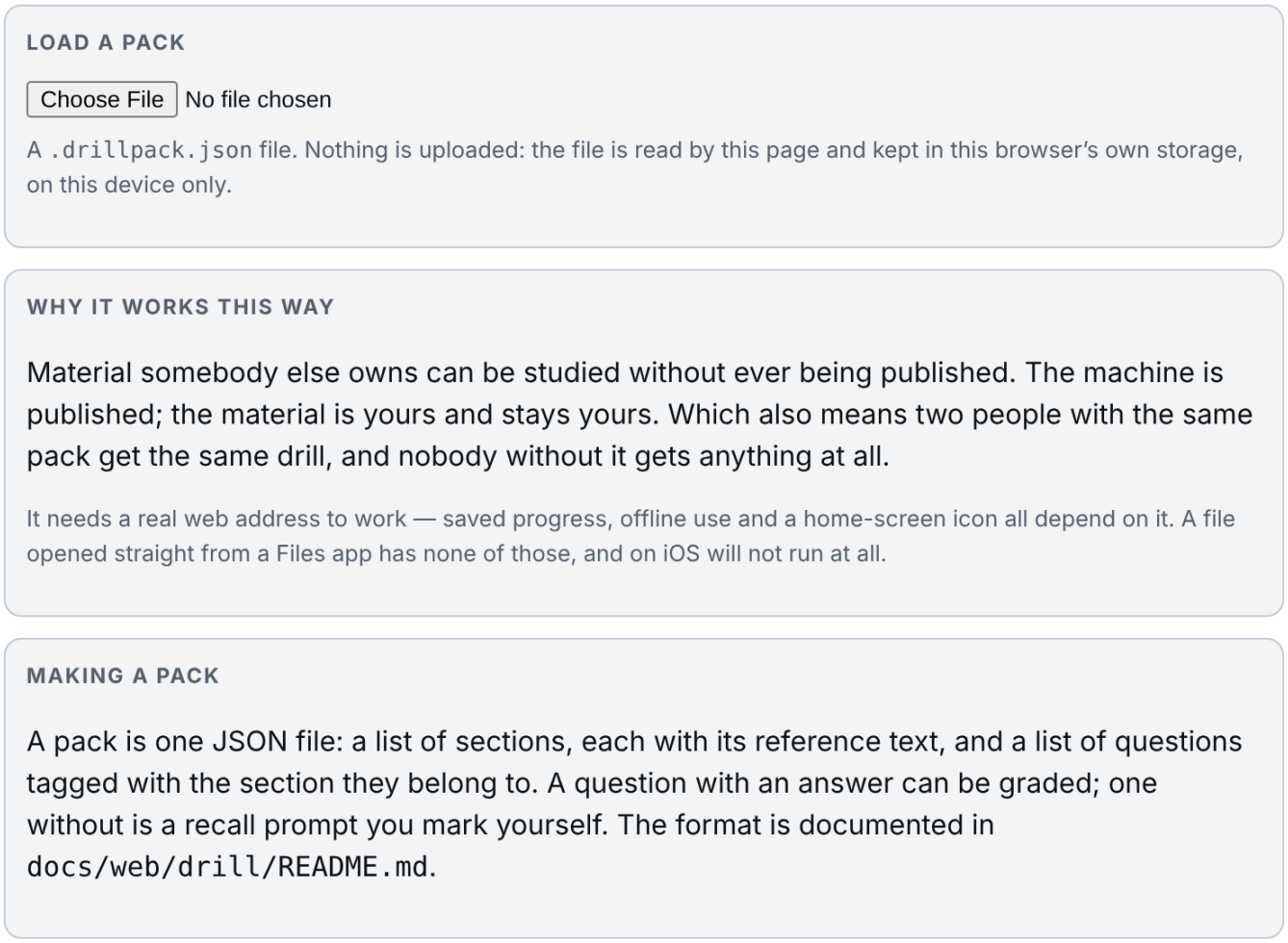



**Figure 1** — what a new copy looks like: the machine, and nowhere for material to have come from.

## 2. How you study with it

| | |
|---|---|
| **The reference** | the pack's own text, always open — including for sections you have not reached |
| **Practice** | recall prompts: the question with no choices, and you check yourself |
| **The test** | graded questions, marked, with the answer and its source |
| **Where am I weak?** | a diagnostic across everything you have worked |

**Sections open in order.** The first is open from the start; each later one opens when the one before it is **passed**. The pass mark is the pack's own, and 80 per cent if the pack does not say.

**The reference is never locked.** Being locked out of a procedure you are carrying this for is the one failure that would matter, so the text is always readable even when the test is not yet open.

**Studied means worked, not opened.** Reading the reference does not count as studying a section; a recall prompt marked or a test taken does.

**Recall first, then the graded ones.** A section's practice run puts the author's own recall prompts first, because they are the harder kind, and then asks the graded questions as recall rather than as a list of choices.

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## 3. The content pack

A pack is a single JSON file named `*.drillpack.json`. It is loaded from the page, validated whole, and kept in the browser's own database. Loading a pack **replaces** the one that was there.

### The shape of it

```
{
  "format":   "bashnav-drill-pack/1",
  "title":    "Rx72 deck procedures",
  "note":     "shipshape, private",
  "built":    "2026-11-12",
  "pass":     80,
  "sections": [
    { "id": "s1", "title": "Reefing", "html": "<p>The reference text…</p>" }
  ],
  "questions": [
    {
      "section": "s1",
      "q":  "What is the first action on the call to reef?",
      "a":  "Ease the kicker and take the weight on the topping lift.",
      "ch": ["Drop the halyard", "Bear away to dead downwind"],
      "where": { "part": "Deck", "sub": "Reefing", "row": "3" },
      "kind": "hand"
    }
  ],
  "back": "<p>Anything to print after the last section.</p>"
}
```

### Every field

#### The pack

| field | required | |
|---|---|---|
| `format` | **yes** | must begin `bashnav-drill-pack/`; anything else is refused as not a pack |
| `sections` | **yes** | a non-empty list |
| `questions` | **yes** | a non-empty list |
| `title` | no | shown at the top; “Content pack” if absent |
| `note` | no | a line under the title — who it is from, whether it is private |
| `built` | no | the pack’s own date, shown beside the app’s build. **Worth setting:** it is how you tell a rebuilt pack from the one it replaced |
| `pass` | no | the pass mark as a percentage; 80 if absent |
| `back` | no | HTML printed after the last section in the reference |

### A section

| field | required | |
|---|---|---|
| `id` | **yes** | unique within the pack; questions refer to it |
| `title` | **yes** | |
| `html` | no | the reference text for this section |

### A question

| field | required | |
|---|---|---|
| `section` | **yes** | must be the `id` of a section in this pack |
| `q` | **yes** | the prompt |
| `a` | no | the answer. **Its presence is what makes a question graded;** a question with no `a` is a recall prompt, and you mark yourself against the reference |
| `ch` | no | wrong options to offer alongside the answer. None may equal `a`, and none may repeat |
| `seq` | no | an ordering question: the steps in their right order, three or more, none repeated |
| `extra` | no | required when `seq` is present: one or more steps that do **not** belong, none of which may also be in `seq` |
| `where` | no | the citation shown under the question: `{ "part": …, "sub": …, "row": … }` |
| `kind` | no | `"hand"` marks a question written by hand rather than generated, and it is labelled as such when asked |

### What a pack is refused for

The pack is validated **whole** and rejected whole, because loading half of one gives a drill with silent holes in it. The reasons, in the tool's own words:

1. *that file does not declare itself a drill pack* — the `format` string.
2. *the pack has no sections* / *no questions*.
3. *section N has no id or title*, or *two sections share the id X*.
4. *N questions belong to sections the pack does not contain*.
5. *question N offers its own answer as a wrong option* — that is a question with two right answers, and the mark it gives is a lie either way. Cheap to check here; impossible to notice while drilling.
6. *question N offers the same wrong option twice*.
7. *question N has fewer than three steps to order*, *lists the same step twice*, *has no step that does not belong*, or *offers a real step as the one that does not belong*.

### Writing a pack

1. **Generate it.** A pack is data; hand-maintained JSON drifts from whatever the questions came out of. Keep the generator with the source material.

2. **Set `built`.** Without it, a rebuilt pack is indistinguishable from the one it replaced, and a question you were sure you had fixed comes round again.
3. **Cite with `where`.** A question whose answer cannot be traced back to the material is a question nobody can settle an argument about.
4. **Keep the reference complete.** It is the part that is always open, and the part that is actually carried.

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## 4. Where the material lives, and who writes packs

The machine is published at `bashnav.navigationtools.org/web/drill/`. Packs are not published with it. Three kinds are expected:

1. **Public packs** to go with the Practical Navigator's lessons.
2. **Private packs** for a particular boat's equipment, never published.
3. **Material owned by somebody else**, kept wherever its owner wants it kept and loaded by the person entitled to it.

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## 5. Offline, and putting it on the home screen

1. Open it once with a signal.
2. **Share → Add to Home Screen.**
3. It runs with the antenna down. The pack and your progress are in the browser's own database on that device.

**A pack is not a backup.** It lives in one browser on one device. Keep the file itself somewhere you can load it again.

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## 6. The display modes

## The display: day, dark and night

The browser build has three display modes. They are not decoration: one of them exists to protect your night vision, and choosing the wrong one at the wrong moment costs you something real.

| Mode | What it is for |
|---|---|
| **NIGHT** | Red on black, for a dark-adapted eye. The default. |
| **DARK** | Near-black with light text, for below decks, dusk, or indoors. |
| **DAY** | White with near-black text, for daylight and for printing. |

### Why night is the default

A tool you open on deck in the dark must not start bright. A bright panel costs a dark-adapted eye roughly **twenty minutes** to recover from, and twenty minutes is a long time to be unable to see a mark or a ship's lights properly. So the first time you open any of these tools you get night, and from then on you get whatever you last chose.

### Three buttons, not a cycle

The three modes are three separate buttons, and the one you are in is shown as selected. That is deliberate.

1. **A cycle can take you somewhere you did not intend.** One stray press with a cycling control can land you on DAY at three in the morning.
2. **A cycle cannot tell you where you are**, only what the next press would do. Three buttons show the current mode at a glance without changing it.
3. **Going darker is always one tap.** DARK and NIGHT are immediate from anywhere, because going darker never costs you anything.
4. **Going from NIGHT to DAY takes a deliberate half-second press.** Hold the DAY button and it fills as it goes; let go early and nothing happens. Nothing bright appears on the screen until the change actually completes. That one transition is guarded because that one transition is the only one that costs you your dark adaptation.

### The setting follows you

The mode is remembered for all of the tools together, and for the course pages that embed them. Set night in one and you are in night in the next. It is remembered in the browser, so it survives closing the app.

### It is right before the page draws

The mode is set in the first instructions the page runs, before any of it is painted. This matters on a first visit over a poor connection: set it later and the page would appear in one palette and then jump to another, which at night is the difference between keeping your night vision and losing twenty minutes of it. If nothing has been stored yet, or whatever was stored cannot be read, the page opens in **night**.

### The controls follow the palette too

Sliders, scroll bars and the other controls the browser draws are told which world they are in, so they are drawn in the mode you chose rather than the browser's own light default. Without that, a night screen can carry a bright white slider across the middle of it.

### If you use High Contrast Mode

On Windows, High Contrast replaces every colour on the page with the ones your system theme specifies. That applies here too: the tools do not override it, because it is your setting and you chose it.

Nothing is lost by it. Where a colour carries meaning — the navigation lamps in colregs — the letter beside it says the same thing, so a lamp is still identifiable when the hue has gone.

**One thing to know if you need both.** High Contrast wins over night mode. If you need night vision on deck *and* high contrast to read, choose a dark high-contrast theme in Windows; the tools will follow it.

### What the display can and cannot do in sunlight

Worth knowing before you blame the app on a bright day.

In sunlight, light reflecting off the glass lands on the dark parts of the screen and the light parts alike. That lifts everything together and flattens the difference between them, so the contrast you actually see collapses no matter what colours the page uses:

| Condition | Body text, as seen |
|---|---|
| Indoors | about 10:1 |
| Overcast on deck | about 6:1 |
| Bright | about 3.5:1 |
| Full sun on the glass | about 2.3:1 |

**At the extreme, every colour scheme converges.** Pure black on pure white measures the same 2.3:1 in full sun as the day mode does, because the glare swamps the difference. A different palette is not the answer, and nor is a different shade of white: across every plausible choice of background the contrast changes by under 4%, which is below what an eye can see.

**What does work, in order:**

1. **Turn the screen brightness up.** This is the one large lever, and it is the device’s, not the app’s.
2. **Shade the screen** with a hand, your body or the sprayhood. Removing the reflection is worth more than any change the app can make.
3. **Use DAY mode**, which at least puts the maximum luminance the panel can give behind the darkest text it can draw.

### A note for printing

Documents and their figures are made in **day** mode. A screen captured in dark or night mode prints as a solid block of toner and loses the fine type in a working, so the printed pages show the day display even if that is not the one you use at sea.

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## 7. Cautions

1. **The pack is the authority, and this tool is not.** A graded answer is the pack author’s answer. Where a pack and the real procedure differ, the procedure is right.
2. **Your progress is per device and per browser.** Clearing site data clears it, and it does not follow you to another device.
3. **Loading a pack replaces the one held.** Progress is kept per section id, so a rebuilt pack that keeps its ids keeps your progress — which is what you want, and also why `built` matters when the questions have changed.

4. **Do not load a pack you were not given.** The material is somebody’s property and this tool makes no claim on it either way.